Mina Tahmasbi Arashloo

Cheriton School of Computer Science
University of Waterloo
200 University Avenue West
Waterloo, ON N2L 3G1, Canada

mina.arashloo@uwaterloo.ca
http://mina.arashloo.net

Research Interest

Networked systems, with an emphasis on software-defined and programmable networks.

Education

Princeton University 2014 - 2019
Ph.D. and M.A in Computer Science
Advisor: Jennifer Rexford
Thesis: Stateful Programming of High-Speed Network Hardware

Sharif University of Technology 2010 - 2014
B.Sc. in Computer Engineering (GPA: 19.50/20.00)

Employment

University of Waterloo 2022 - present
Cheriton School of Computer Science
Assistant Professor

Cornell University 2019 - 2022
Department of Computer Science
Presidential Post-Doctoral Fellow
Supervised by Nate Foster and Rachit Agarwal

Microsoft Research Summer 2017
Research Intern

Microsoft Azure Fall 2016
Intern

Honors and Recognitions

Canada Research Chair in "Minimizing Human Error in Operating Modern Networks" (Tier 2) 2024 - present
Heidelberg Laureate Forum Young Researcher 2023
NSERC Discovery Grant Early Career Supplement 2023
Rising Star in Networking and Communications by N^2Women 2021
Best Paper Award at ACM SIGCOMM SOSR 2020
Cornell Presidential Post-Doctoral Fellow 2019 - 2022
ACM SIGCOMM Doctoral Dissertation Award 2019
Siebel Scholar Class of 2019
Microsoft Research Dissertation Grant 2019
EECS Rising Stars in EECS at MIT 2018
School of Engineering and Applied Science Award of Excellence, Princeton University 2017

Grants

Mitacs Accelerate (with AMD) 2025
Sole PI, $39,800

Canada Research Chair (Tier 2) 2024 - 2029
Sole PI, $500,000 in total including $70,000 annually in research funds

NSERC Discovery Grant 2023 - 2028
Sole PI, $31,000 annually

NSERC Discovery Grant Early Career Supplement 2023
Sole PI, $12,500

Publications

A Protocol-Independent Transport Architecture
Kimiya Mohammadtaheri, David Gao, Samuel Zhang, Matthew Chen, Eric Su, Pengyu Ji, Saad Syed, Chris Neely, Mario Baldi, Nachiket Kapre, Mina Tahmasbi Arashloo
Pre-print, Under Submission

High-Level and Target-Agnostic Transport Programs
Pedro Mizuno, Kimiya Mohammadtaheri, Linfan Qian, Joshua Johnson, Danny Akbarzadeh, Chris Neely, Mario Baldi, Nachiket Kapre, Mina Tahmasbi Arashloo
Pre-print, Under Submission

The Hidden Cost of Heterogeneity: Navigating Internal Contention in SoC SmartNICs
Tyler Liu, Mina Tahmasbi Arashloo, Raouf Boutaba
ACM Special Interest Group on Performance Evaluation, **ACM SIGMETRICS 2027**

The Case for Automated Probabilistic Reasoning for Network Performance Analysis
Jamie Kai, Rachee Singh, Mina Tahmasbi Arashloo
ACM SIGCOMM Workshop on Hot Topics in Networks, **HotNets 2026**

Count-Based Abstractions for Performance Verification of Contention Points
Amir Seyhani, Aarti Gupta, David Walker, Mina Tahmasbi Arashloo
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2026**

Shesha: Opportunistic In-network Acceleration of Asynchronous Distributed Reinforcement Learning
Nehal Baganal Krishna, Anam Tahir, Firas Khamis, Mina Tahmasbi Arashloo, Michael Zink, Amr Rizk
ACM Conference on emerging Networking EXperiments and Technologies, **CoNEXT 2026**

Analyzing Symbolic Properties for DRL Agents in Systems and Networking
Mohammad Zangooei, Jannis Weil, Amr Rizk, Mina Tahmasbi Arashloo, Raouf Boutaba
ACM Special Interest Group on Performance Evaluation, **ACM SIGMETRICS 2026**

Dynamic SLA-Aware Network Slice Monitoring
Niloy Saha, Mina Tahmasbi Arashloo, Nashid Shahriar, Raouf Boutaba
ACM Special Interest Group on Performance Evaluation, **ACM SIGMETRICS 2026**

Toward eBPF-Accelerated Pub-Sub Systems
Beihao Zhou, Samer Al-Kiswany, Mina Tahmasbi Arashloo
ACM SIGCOMM eBPF Workshop, **eBPF 2025**

Buffy: A Formal Language-Based Framework for Network Performance Analysis
Amir Seyhani, Junyi Zhao, Aarti Gupta, David Walker, Mina Tahmasbi Arashloo
ACM SIGCOMM Workshop on Hot Topics in Networks, **HotNets 2024**

Formal Methods for Network Performance Analysis
Mina Tahmasbi Arashloo, Ryan Beckett, Rachit Agarwal
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2023**

dcPIM: Near-Optimal Proactive Datacenter Transport
Qizhe Cai, Mina Tahmasbi Arashloo, Rachit Agarwal

ACM Special Interest Group on Data Communication, **SIGCOMM 2022**

Modular Switch Programming under Resource Constraints
Mary Hogan, Shir Landau-Feibish, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2022**

Towards Formally Verifying Congestion Control Behavior
Venkat Arun, Mina Tahmasbi Arashloo, Ahmed Saeed, Mohammad Alizadeh, Hari Balakrishnan
ACM Special Interest Group on Data Communication, **SIGCOMM 2021**

DBVal: Validating P4 Data Plane Runtime Behavior
K Shiv Kumar, Ranjitha K, P S Prashanth, Mina Tahmasbi Arashloo, Venkanna U., Praveen Tammana
ACM SIGCOMM Symposium on SDN Research, **SOSR 2021**

Petr4: Formal Foundations for P4 Data Planes
Ryan Doenges, Mina Tahmasbi Arashloo, Santiago Bautista, Alexander Chang, Newton Ni, Samwise Parkinson, Rudy Peterson, Alaia Solko-Breslin, Amanda Xu, Nate Foster
ACM SIGPLAN Symposium on Principles of Programming Languages, **POPL 2021**

Elastic Switch Programming with P4All
Mary Hogan, Shir Landau-Feibish, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker, Rob Harrison
ACM Workshop on Hot Topics in Networks, **HotNets 2020**

Enabling Programmable Transport Protocols in High-Speed NICs
Mina Tahmasbi Arashloo, Alexey Lavrov, Manya Ghobadi, Jennifer Rexford, David Walker, David Wentzlaff
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2020**

Tracking P4 Program Execution Path in the Data Plane
Suriya Kodeswaran, Mina Tahmasbi Arashloo, Praveen Tammana, Jennifer Rexford
ACM SIGCOMM Symposium on SDN Research, **SOSR 2020 (Best Paper Award)**

A Scalable VPN Gateway for Multi-Tenant Cloud Services
Mina Tahmasbi Arashloo, Pavel Shirshov, Rohan Gandhi, Guohan Lu, Lihua Yuan, Jennifer Rexford
ACM SIGCOMM Computer Communication Review, **SIGCOMM CCR 2018**

HotCocoa: Hardware Congestion Control Abstractions
Mina Tahmasbi Arashloo, Monia Ghobadi, Jennifer Rexford, David Walker
ACM Workshop on Hot Topics in Networks, **HotNets 2017**

SNAP: Stateful Network-wide Abstractions for Packet Processing
Mina Tahmasbi Arashloo, Yaron Koral, Michael Greenberg, Jennifer Rexford, David Walker
ACM Special Interest Group on Data Communication, **SIGCOMM 2016**

Compiling Path Queries
Srinivas Narayana, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2016**

Teaching

University of Waterloo

CS 456/656: Computer Networks	Fall 2026
CS 856: Network Transport Protocols	Winter 2026
CS 456: Computer Networks	Fall 2025
CS 856: Programmable Networks	Winter 2025
CS 456: Computer Networks	Fall 2024
CS 856: Programmable Networks	Winter 2024
CS 456: Computer Networks	Fall 2023
CS 856: Programmable Networks	Winter 2023

Cornell University

Online certificate series on Software-Defined Networking (SDN)	2020 - 2022

In collaboration with Nate Foster and Cornell Online Education Programs (eCornell)

Professional Service

Program Co-Chair

P4 Workshop	2023
ACM SIGCOMM Symposium on SDN Research (SOSR)	2022
The N^2Women Workshop at SIGCOMM	2022

Organizing Committee

ACM SIGCOMM Non-Paper Session Co-Chair	2026
N2Women Workshop Steering Committee	2025-present
IEEE CNSM Keynote Co-Chair	2023
ACM SIGCOMM Publication co-chair	2020

Program Committee Member

ACM Special Interest Group on Data Communication (SIGCOMM)	2026
USENIX Symposium on Networked Systems Design and Implementation (NSDI)	2026
ACM Special Interest Group on Data Communication (SIGCOMM)	2025
ACM Special Interest Group on Data Communication (SIGCOMM)	2024
USENIX Symposium on Networked Systems Design and Implementation (NSDI)	2024
ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT)	2024
ACM Special Interest Group on Performance Evaluation (SIGMETRICS)	2024
ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets)	2024
USENIX Symposium on Networked Systems Design and Implementation (NSDI)	2023
ACM Symposium on Cloud Computing (SoCC)	2022
ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT)	2021
ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets)	2021
ACM/IEEE Symposium on Architectures for Networking and Communication Systems (ANCS)	2021
ACM Special Interest Group on Data Communication (SIGCOMM)	2020
ACM SIGCOMM Symposium on SDN Research (SOSR)	2019 - 2022
ACM SIGCOMM Posters and Demos	2020
Asia-Pacific Workshop on Networking (APNet)	2020, 2022
EuroP4 Workshop	2020, 2021
P4 Workshop	2018, 2019, 2021

Other Committees and Panels

IRTF Applied Networking Research Prize	2025, 2026
NSF CNS Panel	2024
NSF CISE Panel	2023
NSF CNS Panel	2020

External Reviewer for Conference

ACM SIGPLAN ASPLOS (External Review Committee)	2021
IEEE International Conference on Computer Communications	2017

External Reviewer for Journal

IEEE/ACM Transactions on Networking (TON)	2018 - 2022
IEEE Transactions on Network and Service Management (TNSM)	2021
IEEE Transactions on Very Large Scale Integration Systems (TVLSI)	2018
Journal of Cloud Computing	2017

University Service

Women in Computer Science (WiCS) committee	August 2022 - present

Student Supervision and Mentoring

Graduate Students at University of Waterloo

Anmol Ghadia, MMath
Ryan Xue, MMath
Nesa Abbasimoghadamniasar, MMath
Amir Hossein Seyhani, Ph.D.
Kimiya MohammadTaheri, MMath → Ph.D
Pedro Mizuno, MMath → Ph.D.

Undergraduate Students

Full Time

Andrew Qiu, Undergraduate Researcher	Spring 2026
Pradhyumnan Varadhan, Undergraduate Research Fellow, UBC	Fall 2024, Spring 2025
Linfan Qian, Undergraduate Research Fellow	Fall 2024
Jamie Kai, Undergraduate Research Fellow	Fall 2024
Luc Edes, Undergraduate Research Fellow	Winter 2024

Part Time

Andrew Qiu, Undergraduate Research Assistant	Winter 2026
Matthew Chen, Undergraduate Research Assistant	Winter 2026
Alex Xue, Undergraduate Research Assistant	Fall 2025
Beihao Zhou, Undergraduate Research Assistant (with Samer Al-Kiswany)	Winter 2024, Spring 2025
Joshua Johnson, Undergraduate Research Assistant	Winter 2025
Linfan Qian, Undergraduate Research Assistant	Winter 2025
Zening Hao, Undergraduate Research Assistant	Winter 2025
Joshua Johnson, Undergraduate Research Assistant	Spring 2024
Alex Lin, Undergraduate Research Assistant	Spring 2024
Patrick He, Undergraduate Research Assistant	Spring 2024
Danny Akbarzadeh, Undergraduate Research Assistant	Spring 2024
Carol Duan, Undergraduate Research Assistant	Winter 2024
Crystal Zhou, Undergraduate Research Assistant	Winter 2024
Zening Hao, Undergraduate Research Assistant	Winter 2024
Arnav Nagpal, Undergraduate Research Assistant	Winter 2024
Jiashuo Shi, Undergraduate Research Assistant	Winter 2024
Harshvardhan Kakar, Undergraduate Research Assistant	Spring 2023
Andrew Zang, Undergraduate Research Assistant	Spring 2023
Harshvardhan Kakar, Undergraduate Research Assistant	Spring 2023
Dao Bo Yang, Undergraduate Research Assistant	Winter 2023
Ruiling Ma, Undergraduate Research Assistant	Winter 2023
Kevin Bian, Undergraduate Research Assistant	Winter 2023
Eric Liu, Undergraduate Research Assistant	Fall 2022

Mentorship at Other Institutions

Qizhe Cai, Ph.D. student at Cornell University	2019-2022
Suriya Kudeswaran, M.Sc. student at Princeton University	2018-2019
Melissa Ginaldi, Undergraduate student at Cornell University	2019-2020

Outreach

Co-Organizer of the SmartNIC programming tutorial at ACM SIGCOMM	2026
Panelist, "Thesis Writing and Job/Scholarship Applications", UWaterloo's Women in Math	2025
Panelist, "International STEM Faculty Careers", Princeton University's GradFUTURES	2025
Panelist, "Student & Early Career Networking Researcher Roundtable", the Networking Channel	2023
Co-Organizer of the "Introduction to Network Verification" tutorial at ACM SIGCOMM	2021
Moderator of the "Women at Microsoft" panel at ACM SIGCOMM	2021

Panelist on the "Women in STEM" panel, Rutgers University - Newark Chemistry Society 2021

Invited Talks

IEEE HPSR, Keynote June 2026
Toward an Agile Transport Layer for Programmable Network Hardware

P4 Workshop, Keynote October 2025
High-Level and Target-Agnostic Transport Programs

University of California at Berkeley, System Seminar October 2025
High-Level and Target-Agnostic Transport Programs

IEEE NetSoft, Keynote June 2025
Efficient Abstractions for Network Quality of Service

University of Windsor, School of Computer Science Colloquium Series February 2025
Towards automated proactive analysis of network performance

Dagstuhl Seminar on Network Calculus April 2024
Logic-Based Analysis of Network Performance

ETH (Swiss Federal Institute of Technology) Zurich May 2023
Formal Methods for Network Performance Analysis

Networking and Distributed Systems Lab at HP Labs May 2023
Formal Methods for Network Performance Analysis

USENIX Symposium on Networked Systems Design and Implementation (NSDI) April 2023
Formal Methods for Network Performance Analysis

University of Wisconsin Internet and Systems Research Talk Series April 2021
Enabling Programmable Transport Protocols on High-Speed NICs

Rutgers CS Systems Research Seminar November 2020
Enabling Programmable Transport Protocols on High-Speed NICs

USENIX Symposium on Networked Systems Design and Implementation (NSDI) February 2020
Enabling Programmable Transport Protocols on High-Speed NICs

Guest lecture at the Computer Networks course at MIT November 2019
An Introduction to SmartNICs and their Use Cases

Women Advancing Research, Princeton SheRoars Conference October 2018
Flexible Network Monitoring and Control

ACM Workshop on Hot Topics in Networks November 2017
HotCocoa: Hardware Congestion Control Abstractions

University of Washington Systems Seminar July 2017
SNAP: Stateful Network-wide Abstractions for Packet Processing

Stanford University Networking Seminar May 2017
SNAP: Stateful Network-wide Abstractions for Packet Processing

Network Programming Initiative (NPI) Webinar Series March 2017
SNAP: Stateful Network-wide Abstractions for Packet Processing